

Matrice requisiti - implementazione

Requisito docente	Implementazione	File principali	Demo
Giocatori	CLIENT	users, client pages	login client
Admin locale	LOCAL_ADMIN	user/device/game controllers	login localadmin
Admin gioco	GAME_ADMIN	gameTypeController	login gameadmin
Admin piattaforma	PLATFORM_ADMIN	locale/user/tournament controllers	login platform
Tipi e regole gioco	game_types	game-admin-dashboard	crea tipo
Sensori associati	sensor_templates + sensors	deviceController	crea sensore
Edge locale	edge-service	simulator.js	porta 8090
Offline	coda JSON	simulator/data	pulsante offline
MQTT	Mosquitto QoS 1	mqttClient.js	simula partita
Attuatori	actuators	actuatorService.js	display stato
REST	API gateway e servizi	routes + openapi	browser/Postman
Microservizi	3 servizi + gateway	docker-compose	docker ps
Partite individuali	participant_mode	matchController	start individual
Partite a squadre	teams/team_members	teams page	start TEAM
Tornei multi-locale	tournament_locations	tournaments page	seleziona locali
Calendario/turni	round_number/scheduled_at	tournament_matches	collega partita
Classifica	tournamentRules	tournament detail	vedi punti
Statistiche	statsController	dashboard	statistiche
UML e sequenze	Mermaid	docs/03 e docs/14	relazione
OpenAPI	YAML completo	docs/openapi.yaml	Swagger Editor
Test	Node test	backend/tests	npm test